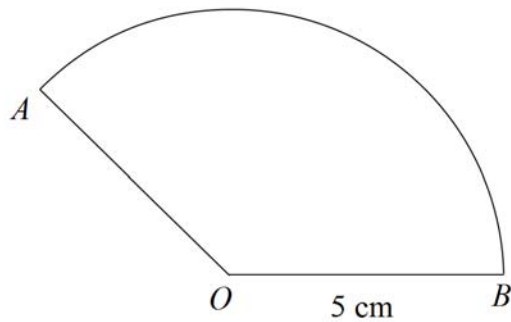


Section I**10 marks****Use the multiple choice answer sheet for questions 1-10.****Marks**

- 1 What is the derivative of $f(x) = \sqrt{3-x}$? 1
- (A) $f'(x) = \frac{-1}{2\sqrt{3-x}}$
- (B) $f'(x) = \frac{-2}{\sqrt{3-x}}$
- (C) $f'(x) = \frac{2}{\sqrt{3-x}}$
- (D) $f'(x) = \frac{1}{2\sqrt{3-x}}$
- 2 The gradient of the normal to the curve $f(x) = x^2 - 4x$ at $(1, -3)$ is 1
- (A) -10
- (B) -2
- (C) $-\frac{1}{2}$
- (D) $\frac{1}{2}$
- 3 Which of the following is true for the equation $3x^2 - x - 2 = 0$? 1
- (A) No real roots
- (B) Equal roots
- (C) Two real distinct roots
- (D) Irrational roots
- 4 The exact value of $\cos \frac{2\pi}{3}$ is? 1
- (A) $-\frac{\sqrt{3}}{2}$
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) $\frac{\sqrt{3}}{2}$

Marks

- 5 AOB is a sector of a circle, centre O and radius 5 cm.
The sector has an area of 10π .

1

Not to scale

What is the arc length of the sector?

- (A) 2π
(B) 4π
(C) 6π
(D) 10π
- 6 What is the derivative of $\log_2 x$?

1

- (A) $\frac{1}{x}$
(B) $\frac{1}{2x}$
(C) $\ln 2x$
(D) $\frac{1}{x \ln 2}$

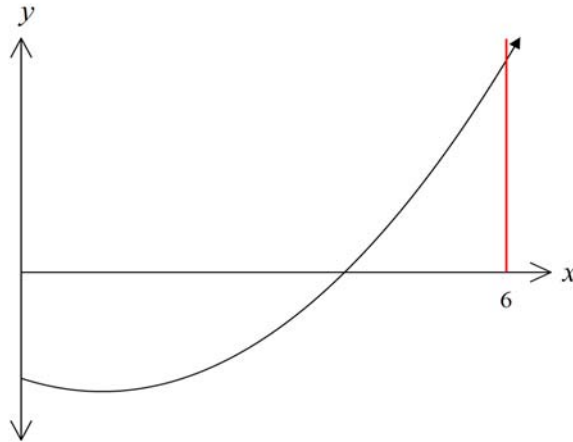
- 7 What is the value of $\int_{-1}^2 x^2 + 1 dx$?

1

- (A) 4
(B) 5
(C) 6
(D) 7

Marks

- 8 The diagram below shows the graph of $y = x^2 - 2x - 8$.

1

What is the correct expression for the area bounded by the x -axis and the curve $y = x^2 - 2x - 8$ between $0 \leq x \leq 6$?

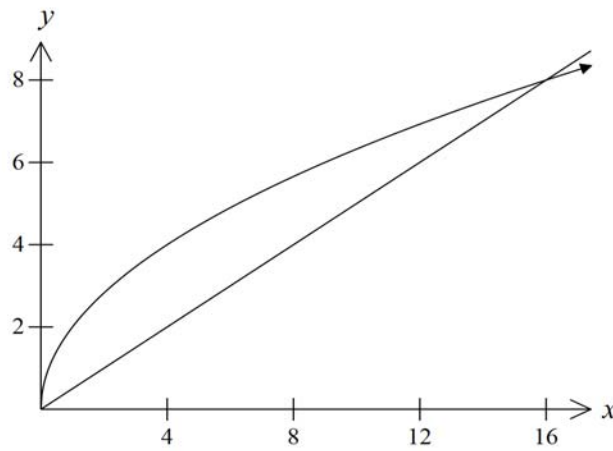
- (A) $A = \int_0^5 x^2 - 2x - 8 dx + \left| \int_5^6 x^2 - 2x - 8 dx \right|$
- (B) $A = \int_0^4 x^2 - 2x - 8 dx + \left| \int_4^6 x^2 - 2x - 8 dx \right|$
- (C) $A = \left| \int_0^5 x^2 - 2x - 8 dx \right| + \int_5^6 x^2 - 2x - 8 dx$
- (D) $A = \left| \int_0^4 x^2 - 2x - 8 dx \right| + \int_4^6 x^2 - 2x - 8 dx$
- 9 What are the solutions to the equation $e^{6x} - 7e^{3x} + 6 = 0$?

1

- (A) $x = 1$ and $x = 6$
- (B) $x = 0$ and $x = \frac{\ln 6}{2}$
- (C) $x = 0$ and $x = \frac{\ln 6}{3}$
- (D) $x = 1$ and $x = \frac{\ln 6}{2}$

Marks

- 10 The diagram below shows the graph of $y = 2\sqrt{x}$ and $y = \frac{x}{4}$.

1

Which of the following is the correct expression for the volume of the solid of revolution when the area between the curve $y = 2\sqrt{x}$ and $y = \frac{x}{4}$ is rotated around the x -axis?

- (A) $V = \int_0^8 (4y - \frac{y^2}{2}) dy$
- (B) $V = \int_0^{16} (2\sqrt{x} - \frac{x}{4}) dx$
- (C) $V = \pi \int_0^8 (16y^2 - \frac{y^4}{4}) dy$
- (D) $V = \pi \int_0^{16} (4x - \frac{x^2}{16}) dx$

End of Section I

Section II

Answer questions in the writing booklet provided.

Start each question in a new booklet. Extra writing booklets are available.

Question 11 (15 marks) Start a new booklet.

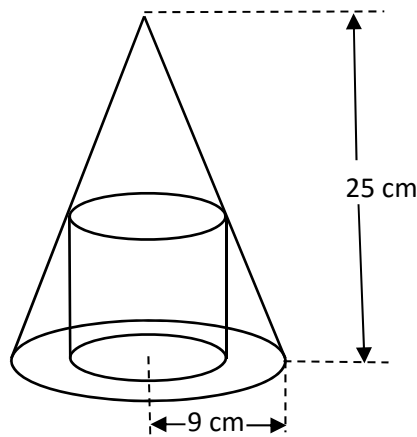
Marks

- (a) Consider the parabola $x^2 - 6x + 4y + 17 = 0$
- (i) Express the parabola in the form $(x - h)^2 = 4a(y - k)$ **1**
 - (ii) Write down the co-ordinates of the vertex. **1**
 - (iii) Write down the coordinates of the focus. **1**
 - (iv) Write down the equation of the directrix. **1**
- (b) If α and β are the roots of the equation $3x^2 + 15x + 6 = 0$, find the value of
- (i) $\alpha + \beta$ **1**
 - (ii) $\alpha\beta$ **1**
 - (iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ **2**
- (c) If $A(x-1)^2 + B(x-1) + C = 3x^2 - 5x + 7$, find A, B and C . **3**
- (d) Find the value of k for which the equation $x^2 + (k+1)x + (k+3) = 0$ has one root equal to 4. **1**
- (e) Derive the equation of the locus of the point $P(x, y)$ which moves so that $AP:PB = 2:3$, where A is $(-2, 1)$ and B is $(3, -3)$. **3**

Question 12 (16 marks) Start a new booklet.

- (a) The equation of a curve is given by $y = 6x^3 + 9x^2 - 3$
- (i) Show that $6x^3 + 9x^2 - 3 = 3(x+1)^2(2x-1)$. 1
- (ii) Hence write down the co-ordinates of the x intercepts. 1
- (iii) Find any stationary points and determine their nature. 4
- (iv) Find the co-ordinates of any points of inflexion. 2
- (v) Sketch the curve showing all essential details. 2

- (b) A cylinder is inscribed in a cone of radius 9 cm and height 25cm.



- (i) Show that the height of the cylinder is $h = \frac{25(9-r)}{9}$, 2
- where r is the radius of the cylinder.
- (ii) Show that the volume of the cylinder is $\frac{25\pi r^2(9-r)}{9}$. 1
- (iii) Hence find the maximum volume of the cylinder. 3

Question 13 (15 marks) Start a new booklet.

(a) Find

(i) $\int (3x+4)^4 dx$ 1

(ii) $\int \frac{4x^2+6}{x^2} dx$ 2

(b) Evaluate $\int_0^5 \sqrt{25-x^2} dx$ 2

(c) The derivative of the curve $y = f(x)$ is given by $f'(x) = x^2 - 3x - 4$. 2

Find the equation of the curve, given that the curve passes through the point $(-2, 1)$.

(d) (i) Sketch on the same axes the curves $y = 4 - x^2$ and $y = 2x^2 - 6x + 4$. 2

(ii) Hence find the area bounded by the parabolas $y = 4 - x^2$ and $y = 2x^2 - 6x + 4$. 3

(e) (i) Use the Trapezoidal Rule with 3 function values to find an approximation to $\int_1^3 \log_e x dx$. Answer correct to 2 decimal places. 2

(ii) State whether the approximation found in part (i) is greater than or less than the exact value of $\int_1^3 \log_e x dx$. Justify your answer. 1

Question 14 (15 marks) Start a new booklet.**Marks**

(a) Differentiate

(i) $y = e^{4x+3}$ 1

(ii) $y = x \ln x$ 1

(iii) $y = \log_e(2x+7)$ 1

(iv) $y = \ln\left(\frac{2x+3}{2x-3}\right)$ 2

(b) Find the integral of

(i) $\int \frac{4x}{4x^2+7} dx$ 1

(ii) $\int x^3 e^{x^4+2} dx$ 1

(c) If $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$ find

(i) $\log_{10} 6$ 1

(ii) $\log_{10} 5$ 1

(d) (i) Sketch the graph of $y = \log_e(x-3)$. 1(ii) State its domain and range. 2(iii) The area between the curve $y = \log_e(x-3)$, the y axis and the lines 3

$y=1$ and $y=4$ is rotated about the y axis. Find the exact volume of the solid formed.

End of paper

Section 1

$$\begin{aligned}
 1. \quad f(x) &= \sqrt{3-x} \\
 &= (3-x)^{\frac{1}{2}} \\
 f'(x) &= \frac{1}{2}(3-x)^{-\frac{1}{2}} \times -1 \\
 &= \frac{-1}{2\sqrt{3-x}} \quad \textcircled{A}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad f(x) &= x^2 - 4x \\
 f'(x) &= 2x - 4 \text{ at } x=1 \\
 &= 2(1) - 4 \\
 &= -2 \\
 \text{gradient of tangent} &= -2 \\
 \text{gradient of normal} &= \frac{1}{2} \\
 \text{as } m_1 m_2 &= -1 \text{ for perpendicular lines} \quad \textcircled{D}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad 3x^2 - x - 2 &= 0 \\
 \Delta &= b^2 - 4ac \\
 &= (-1)^2 - 4(3)(-2) \\
 &= 1 + 24 \\
 &= 25
 \end{aligned}$$

As the discriminant is a perfect square, there are two distinct rational roots \textcircled{C}

$$\begin{aligned}
 4. \quad \cos\left(\frac{2\pi}{3}\right) &= \cos\left(\frac{2 \times 180}{3}\right) \\
 &= \cos 120 \\
 &= -\frac{1}{2} \quad \textcircled{B}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad A &= \frac{1}{2} r^2 \theta \\
 10\pi &= \frac{1}{2} \times 5^2 \theta \\
 \theta &= \frac{20\pi}{25} \\
 \therefore \theta &= \frac{4\pi}{5}
 \end{aligned}$$

$$\begin{aligned}
 L &= r\theta \\
 &= 5 \times \frac{4\pi}{5}
 \end{aligned}$$

$$\therefore L = 4\pi \quad \textcircled{B}$$

$\therefore$ Length of the arc is 4π cm

$$\begin{aligned}
 6. \quad y &= \log_2 x \\
 &= \frac{\log_e x}{\log_e 2} \\
 \frac{dy}{dx} &= \frac{\frac{1}{x}}{\log_e 2} \\
 &= \frac{1}{x \ln 2} \quad \textcircled{D}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \int_{-1}^2 x^2 + 1 \, dx &= \left[\frac{x^3}{3} + x \right]_{-1}^2 \\
 &= \left[\frac{2^3}{3} + 2 \right] - \left[\frac{(-1)^3}{3} - 1 \right] \\
 &= 4\frac{2}{3} + 1\frac{1}{3} \\
 &= 6
 \end{aligned}$$

$$\begin{aligned}
 8. \quad y &= x^2 - 2x - 8 \\
 &= (x-4)(x+2)
 \end{aligned}$$

$$\therefore A = \left| \int_0^4 x^2 - 2x - 8 \, dx \right| +$$

$$\int_4^6 x^2 - 2x - 8 \, dx \quad \textcircled{D}$$

$$9. e^{6x} - 7e^{3x} + 6 = 0$$

$$\text{Let } y = e^{3x}$$

$$y^2 = (e^{3x})^2$$

$$= e^{6x}$$

$$\therefore y^2 - 7y + 6 = 0$$

$$(y-6)(y-1) = 0$$

$$y = 6, 1$$

$$\text{But } y = e^{3x}$$

$$e^{3x} = 1 \quad e^{3x} = 6$$

$$\therefore x = 0 \quad \ln e^{3x} = \ln 6$$

$$3x \ln e = \ln 6$$

$$x = \frac{\ln 6}{3}$$

$$\therefore x = 0, x = \frac{\ln 6}{3} \quad (C)$$

$$10 \quad y = 2\sqrt{x} \quad y = \frac{\pi}{4}$$

$$y^2 = 4x \quad y^2 = \frac{\pi^2}{16}$$

$$V = \pi \int_a^b y^2 dx$$

$$= \pi \int_0^{\frac{\pi^2}{16}} \left(4x - \frac{x^2}{16} \right) dx \quad (D)$$

Section II

$$a) x^2 - 6x + 4y + 17 = 0$$

$$x^2 - 6x = -4y - 17$$

$$x^2 - 6x + \left(\frac{-6}{2}\right)^2 = -4y - 17 + \left(\frac{-6}{2}\right)^2$$

$$(x-3)^2 = -4y - 8$$

$$(x-3)^2 = -4(y+2)$$

$$(x-h)^2 = -4a(y-k)$$

$$a = 1$$

$$ii) \therefore \text{vertex is } (3, -2)$$

$$iii) \therefore \text{focus is } (3, -3)$$

$$iv) \text{directrix is } y = -1$$

$$b) 3x^2 + 15x + 6 = 0$$

$$a = 3 \quad b = 15 \quad c = 6$$

$$i) \alpha + \beta = \frac{-b}{a}$$

$$= \frac{-15}{3}$$

$$= -5$$

$$ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{6}{3}$$

$$= 2$$

$$iii) \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(-5)^2 - 2 \times 3}{3}$$

$$= \frac{25 - 6}{3}$$

$$= \frac{19}{3}$$

$$= 6\frac{1}{3}$$

$$c) A(x-1)^2 + B(x-1) + C = 3x^2 - 5x + 7$$

$$A(x^2 - 2x + 1) + Bx - B + C = 3x^2 - 5x + 7$$

$$Ax^2 - 2Ax + A + Bx - B + C = 3x^2 - 5x + 7$$

$$Ax^2 + (B - 2A)x + (A - B + C) = 3x^2 - 5x + 7$$

$$\therefore A = 3 \quad B - 2A = -5 \quad A - B + C = 7$$

$$B - 2(3) = -5 \quad 3 - 1 + C = 7$$

$$B = 1 \quad C = 5$$

$$\therefore A = 3 \quad B = 1 \quad C = 5$$

$$d) x^2 + (k+1)x + (k+3) = 0$$

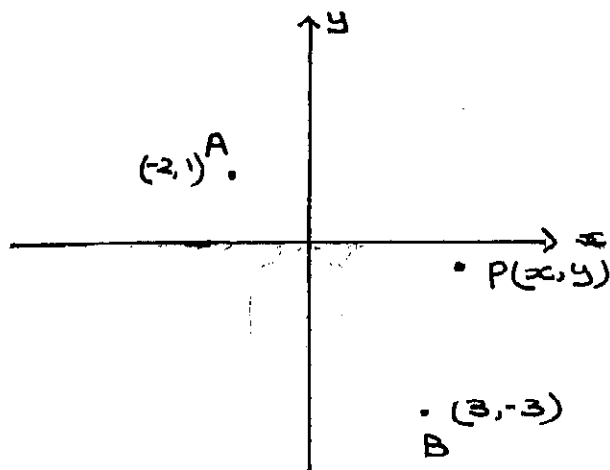
$$4^2 + 4(k+1) + (k+3) = 0$$

$$16 + 4k + 4 + k + 3 = 0$$

$$5k = -23$$

$$k = \frac{-23}{5}$$

c)



$$\frac{AP}{PB} = \frac{2}{3}$$

$$3AP = 2PB$$

$$(3AP)^2 = (2PB)^2$$

$$9AP^2 = 4PB^2$$

Let $P(x, y)$ be a point on the locus

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AP = \sqrt{(x+2)^2 + (y-1)^2}$$

$$AP^2 = (x+2)^2 + (y-1)^2$$

$$PB = \sqrt{(x-3)^2 + (y+3)^2}$$

$$PB^2 = (x-3)^2 + (y+3)^2$$

But

$$9AP^2 = 4PB^2$$

$$9[(x+2)^2 + (y-1)^2] = 4[(x-3)^2 + (y+3)^2]$$

$$9[x^2 + 4x + 4 + y^2 - 2y + 1] = 4[x^2 - 6x + 9 + y^2 + 6y + 9]$$

$$9x^2 + 36x + 9y^2 - 18y + 45 = 4x^2 - 24x + 4y^2 + 24y + 72$$

$$5x^2 + 60x + 5y^2 - 42y - 27 = 0$$

Question 12

$$a) i) y = 6x^3 + 9x^2 - 3$$

$$6x^3 + 9x^2 - 3 = 3(x+1)^2(2x-1)$$

$$R.H.S = 3(x+1)^2(2x-1)$$

$$= 3(x^2 + 2x + 1)(2x - 1)$$

$$= (3x^2 + 6x + 3)(2x - 1)$$

$$= 6x^3 + 12x^2 + 6x - 3x^2 - 6x - 3$$

$$= 6x^3 + 9x^2 - 3$$

$$= L.H.S$$

$$\therefore 6x^3 + 9x^2 - 3 = 3(x+1)^2(2x-1)$$

$$ii) x\text{-intercepts} \Rightarrow y = 0$$

$$6x^3 + 9x^2 - 3 = 0$$

$$3(x+1)^2(2x-1) = 0$$

$$x = -1, \frac{1}{2}$$

$$\therefore x\text{ intercepts are } (-1, 0) \left(\frac{1}{2}, 0\right)$$

$$iii) y = 6x^3 + 9x^2 - 3$$

$$\frac{dy}{dx} = 18x^2 + 18x$$

$$\frac{d^2y}{dx^2} = 36x + 18$$

For stationary points $\frac{dy}{dx} = 0$

$$18x^2 + 18x = 0$$

$$18x(x+1) = 0$$

$$x = 0, -1$$

when $x = 0$

$$y = 6x^3 + 9x^2 - 3$$

$$= -3$$

$\therefore (0, -3)$ is a stationary point

Test the nature

$$\frac{d^2y}{dx^2} = 36x + 18 \text{ at } x=0$$
$$= 18$$

As $\frac{d^2y}{dx^2} > 0$ the curve is concave up $\cup$

$\therefore (0, -3)$ is a local minimum

when $x = -1$

$$y = 6x^3 + 9x^2 - 3$$
$$= 6(-1)^3 + 9(-1)^2 - 3$$
$$= -6 + 9 - 3$$
$$= 0$$

$\therefore (-1, 0)$ is a stationary point

Test the nature

$$\frac{d^2y}{dx^2} = 36x + 18 \text{ at } x = -1$$
$$= 36(-1) + 18$$
$$= -18$$

As $\frac{d^2y}{dx^2} < 0$ the curve is concave down $\cap$

$\therefore (-1, 0)$ is a local maximum.

iv) For points of inflexion

$$\frac{d^2y}{dx^2} = 0 \text{ and there}$$

is a change in concavity

$$\frac{d^2y}{dx^2} = 36x + 18$$

$$36x + 18 = 0$$

$$36x = -18$$

$$x = -\frac{1}{2}$$

when $x = -\frac{1}{2}$

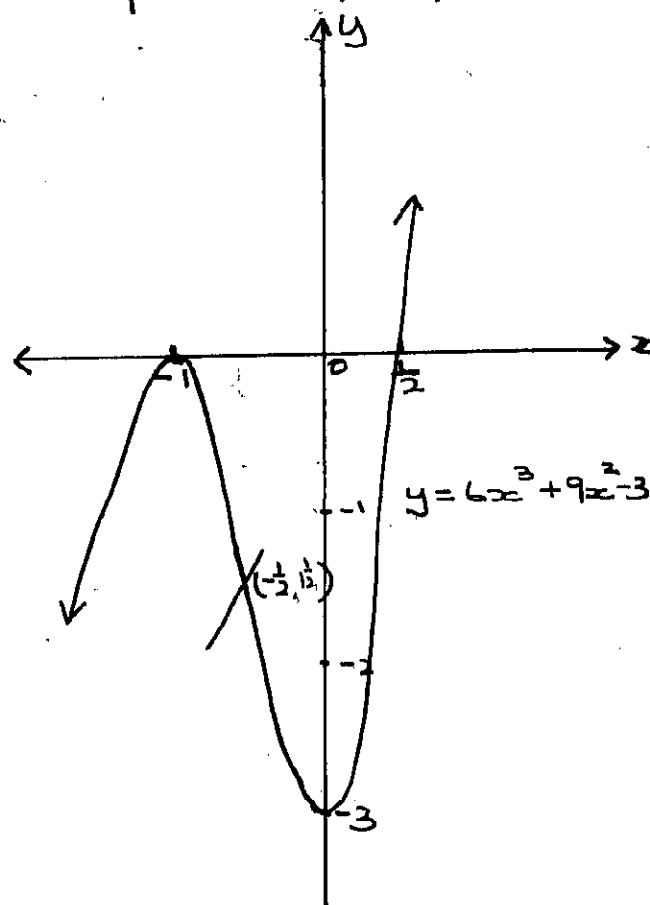
$$y = 6x^3 + 9x^2 - 3$$
$$= 6\left(-\frac{1}{2}\right)^3 + 9\left(-\frac{1}{2}\right)^2 - 3$$
$$= -1\frac{1}{2}$$

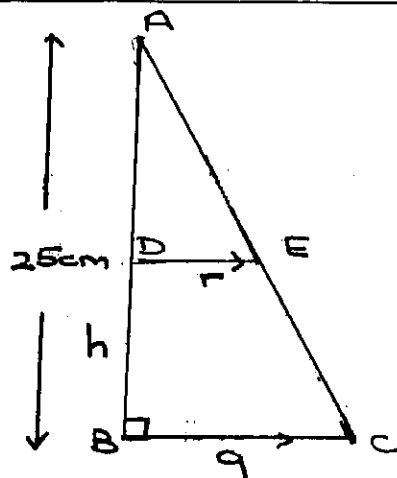
$\therefore \left(-\frac{1}{2}, -1\frac{1}{2}\right)$ may be a point of inflexion.

check concavity

x	$-\frac{1}{2}^-$	$-\frac{1}{2}$	$-\frac{1}{2}^+$
$\frac{d^2y}{dx^2} = 36x + 18$	-	0	+

As there is a change in concavity $\left(-\frac{1}{2}, -1\frac{1}{2}\right)$ is a point of inflexion





In $\triangle ADE$, $\triangle ABC$

$$\hat{ADE} = \hat{ABC}$$

$$= 90^\circ$$

$\hat{AED} = \hat{ACB}$ corresponding angles are equal $DE \parallel BC$

$\therefore \triangle ADE \parallel \triangle ABC$ equiangular

$$\frac{BC}{DE} = \frac{AB}{AD} \text{ corresponding sides of}$$

$$\frac{9}{r} = \frac{25}{25-h} \text{ similar triangles are in proportion}$$

$$9(25-h) = 25r$$

$$225 - 9h = 25r$$

$$9h = 225 - 25r$$

$$= 25(9-r)$$

$$h = \frac{25(9-r)}{9}$$

$$V = \pi r^2 h$$

$$= \pi r^2 \times \frac{25(9-r)}{9}$$

$$= \frac{25\pi r^2 (9-r)}{9}$$

$$V = \frac{225\pi r^2 - 25\pi r^3}{9}$$

$$\frac{dV}{dr} = \frac{450\pi r - 75\pi r^2}{9}$$

$$= \frac{25}{3} \pi r (6-r)$$

For stationary points $\frac{dV}{dr} = 0$

$$\frac{75\pi r (6-r)}{9} = 0$$

$$75\pi r (6-r) = 0$$

$$r = 0, 6 \quad r > 0$$

$$\therefore r = 6$$

when $r = 6$

$$V = \frac{25\pi r^2 (9-r)}{9}$$

$$= \frac{25\pi (6)^2 (9-6)}{9}$$

$$= 300\pi$$

Test the nature

$$\frac{dV}{dr} = \frac{150\pi r - 25\pi r^2}{3}$$

$$\frac{d^2V}{dr^2} = 50\pi - \frac{50\pi r}{3}$$

$$= \frac{50\pi - 50\pi(6)}{3}$$

$$= 50\pi - 100\pi$$

$$= -50\pi$$

As $\frac{d^2V}{dr^2} < 0 \quad \checkmark$ a

maximum value occurs

$\therefore$ The maximum volume is 300π

Question 13

a) (i) $\int (3x+4)^4 dx = \frac{1}{5}(3x+4)^5 + c$

ii) $\int \frac{4x^2+6}{x^2} dx = \int 4 + 6x^{-2} dx$
 $= 4x - 6x^{-1} + c$
 $= 4x - \frac{6}{x} + c$

b) $\int_0^5 \sqrt{25-x^2} dx = \frac{1}{4} \pi r^2$
 $= \frac{1}{4} \pi (5)^2$
 $= \frac{25\pi}{4}$

c) $f'(x) = x^2 - 3x - 4$

$f(x) = \int x^2 - 3x - 4 dx$
 $= \frac{x^3}{3} - \frac{3x^2}{2} - 4x + c$

when $x = -2$, $f(x) = 1$

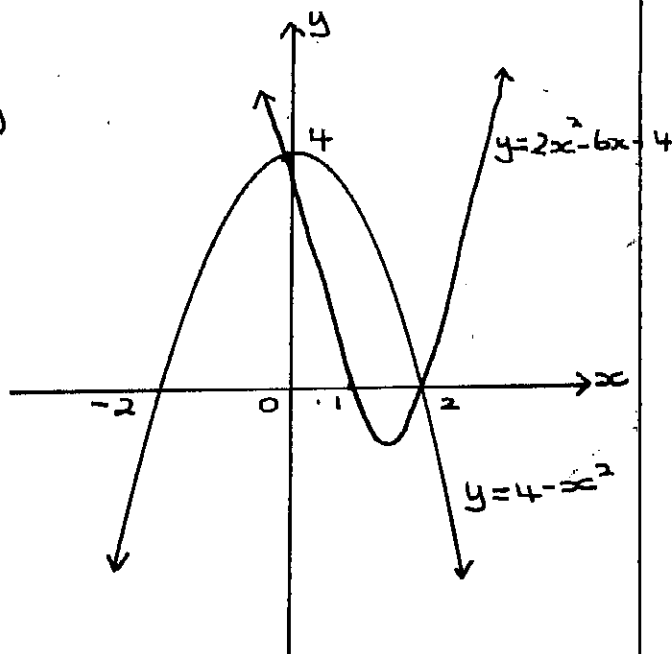
$1 = \frac{(-2)^3}{3} - \frac{3(-2)^2}{2} - 4(-2) + c$

$1 = -\frac{8}{3} - 6 + 8 + c$

$c = \frac{5}{3}$

$\therefore f(x) = \frac{x^3}{3} - \frac{3x^2}{2} - 4x + \frac{5}{3}$

d)



$A = \int_0^2 (4 - x^2 - (2x^2 - 6x + 4)) dx$

$= \int_0^2 (4 - x^2 - 2x^2 + 6x - 4) dx$

$= \int_0^2 (-3x^2 + 6x) dx$

$= \left[-x^3 + 3x^2 \right]_0^2$

$= \left[-2^3 + 3(2)^2 \right] - \left[0 \right]$

$= 4$

$\therefore$ Area is 4 units²

e)

x	1	2	3
$f(x) = \ln x$	0	$\ln 2$	$\ln 3$

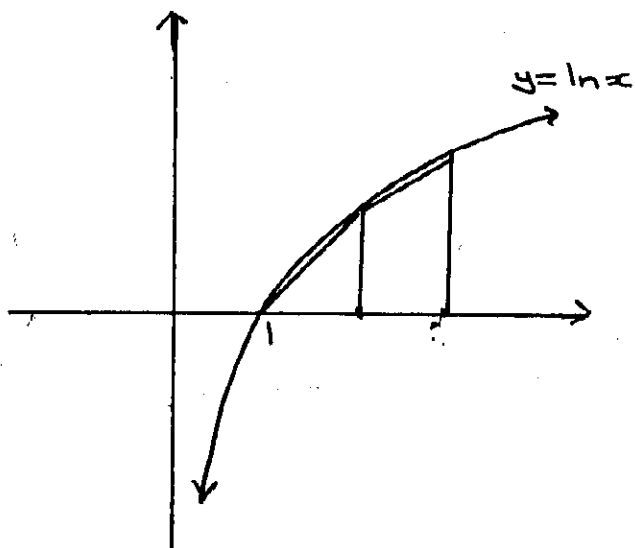
$\int_a^b f(x) dx \doteq \frac{b-a}{2} [f(a) + f(b)]$

$\int_1^3 \ln x dx \doteq \frac{2-1}{2} [0 + 2\ln 2 + \ln 3]$

$\doteq 1.242453325$

$= 1.24$ to 2dp

b)



The area using the trapezoidal rule is less than the area under the curve.
The curve is concave down

Question 14

a) (i) $y = e^{4x+3}$

$$\frac{dy}{dx} = 4e^{4x+3}$$

(ii) $y = x \ln x$ $u = x$ $v = \ln x$
 $= u v$ $\frac{du}{dx} = 1$ $\frac{dv}{dx} = \frac{1}{x}$

$$\begin{aligned} \frac{dy}{dx} &= v \frac{du}{dx} + u \frac{dv}{dx} \\ &= \ln x + x \times \frac{1}{x} \\ &= \ln x + 1 \end{aligned}$$

(iii) $y = \log_e(2x+7)$

$$\frac{dy}{dx} = \frac{2}{2x+7}$$

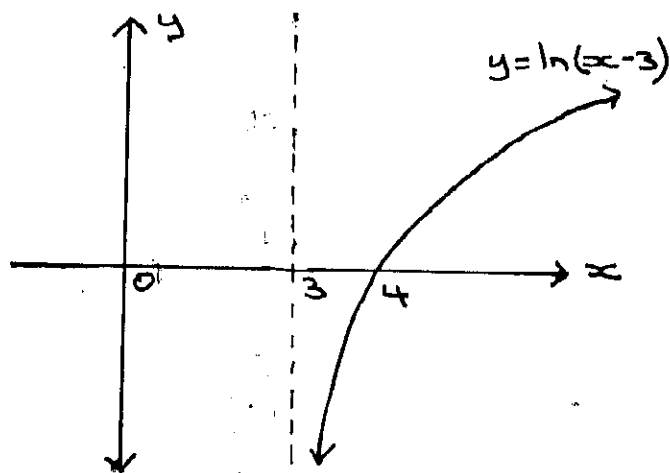
$$\begin{aligned} 1) y &= \ln \left(\frac{2x+3}{2x-3} \right) \\ &= \ln(2x+3) - \ln(2x-3) \\ \frac{dy}{dx} &= \frac{2}{2x+3} - \frac{2}{2x-3} \\ &= \frac{2(2x-3) - 2(2x+3)}{(2x+3)(2x-3)} \\ &= \frac{-12}{4x^2-9} \end{aligned}$$

$$\begin{aligned} b) i) \int \frac{4x}{4x^2+7} dx &= \frac{1}{2} \int \frac{8x}{4x^2+7} dx \\ &= \frac{1}{2} \ln(4x^2+7) + C \end{aligned}$$

$$ii) \int x^3 e^{x^4+2} dx = \frac{1}{4} e^{x^4+2} + C$$

$$\begin{aligned} c) \log_{10} 6 &= \log_{10}(2 \times 3) \\ &= \log_{10} 2 + \log_{10} 3 \\ &= 0.301 + 0.477 \\ &= 0.778 \end{aligned}$$

$$\begin{aligned} ii) \log_{10} 5 &= \log_{10} \left(\frac{10}{2} \right) \\ &= \log_{10} 10 - \log_{10} 2 \\ &= 1 - 0.301 \\ &= 0.699 \end{aligned}$$



domain $x > 3$

range $\mathbb{R}$

$$y = \log_e (x-3)$$

$$e^y = x-3$$

$$x = e^y + 3$$

$$\begin{aligned} x^2 &= (e^y + 3)^2 \\ &= e^{2y} + 6e^y + 9 \end{aligned}$$

$$V = \pi \int_a^b x^2 dy$$

$$= \pi \int_1^4 e^{2y} + 6e^y + 9 dy$$

$$= \pi \left[\frac{1}{2} e^{2y} + 6e^y + 9y \right]_1^4$$

$$= \pi \left[\frac{1}{2} e^8 + 6e^4 + 36 \right] - \pi \left[\frac{1}{2} e^2 + 6e + 9 \right]$$

$$= \pi \left[\frac{1}{2} e^8 + 6e^4 - \frac{1}{2} e^2 - 6e + 27 \right]$$

$$\therefore \text{Volume is } \pi \left[\frac{1}{2} e^8 + 6e^4 - \frac{1}{2} e^2 - 6e + 27 \right] \text{ units}^3$$